



Test Report No. 2024 – 0563



Agricultural Machinery – Knapsack Sprayer SATO PS-20L-TU26

Test Report Number	2024 – 0563
Date of Issuance	November 30, 2024
Valid Until	November 30, 2029
Agricultural Machinery	Knapsack Sprayer
Type	Power-operated, Engine-driven
Brand	SATO
Model	PS-20L-TU26
Test Requested by	MEGA SK INDUSTRIAL CORP. 1 st Floor, Philpremiere Bldg., 10 Niog Road, Brgy. Niog III, Bacoor City, Cavite

SUMMARY

Performance Criteria	Requirement as per PNS/BAFS 331:2022	AMTEC Test
Volumetric Efficiency, %, minimum	80	NA ¹
Pressure Drop During Leakage Test	Shall not drop by more than 50% of the applied pressure	NA ²
Continuous Running Test	Shall not have any abnormality or trouble	No abnormality or trouble
Flow Rate, L/min	0.3-1.40	1.56-6.15
Spray Quality, VMD, µm	Medium (236-340)	Medium (236)

Note: NA- Not Applicable (¹ The knapsack sprayer is engine driven, ² Leakage test was not conducted)

OBSERVATIONS

1. Operator's manual (based on PAES 102:2000)	
a. Language	English
b. Safety and warnings	Provided
c. Operating information	Provided
d. Accessories and attachments	Provided
e. Maintenance instruction	Provided
f. Storage	Provided
g. Handling, reception, transportation, assembly, and installation	Not provided
h. Specifications	Provided
i. Dismantling and disposal	Not provided
j. Warranty	Not provided
k. Parts list	Provided
2. Previous test reports of similar brand and model	None
3. Manufacturer	China
4. Serial number	TMH0720240114
5. Number of operators	One
6. Position and shape of lever (if any) and fatigue in operation	Not applicable; the sprayer is engine-driven.
7. Manipulating cut-off device	The cut-off device is located at the end of the lance which is accessible to the operator.
8. Disassembly or assembly and maintenance	The spray tank, engine, lance, and cut-off device can be detached for maintenance and repair.
9. Spillage at the tank filler cap and cylinder cap	Spillage at the tank filler cap and cylinder cap were observed in horizontal positions during the tilt and inversion test (Figure 3).
10. Dripping on the cutoff valve, lance, and nozzle connections	There were no drippings observed on the cut-off valve, lance, and nozzle connections during the test operation.
11. Conformity to the back of the operator	The operator was able to carry the sprayer on the back (Figure 5).
12. Tank filler cap test	The tank filler cap was not removed when the sprayer was pushed over.
13. Presence of hot surfaces	Hot surfaces of the engine are covered and are not accessible to the operator when the machine is worn.
14. All permanent joints shall not show any sign of failure when tested for durability, as specified in the PNS/BAFS 332:2022 (Knapsack sprayers – Methods of test).	All permanent joints showed no signs of failure when tested for durability.

OBSERVATIONS (Cont'n)

15. Pressurized parts of the sprayer shall withstand twice the rated operating pressure.	The pressure test was not conducted due to the unavailability of test equipment.
16. For engine-driven knapsack sprayer, the tank nominal volume should not be more than 40 kg. For other sprayers (i.e., lever-operated, compression, electric motor-driven, and dual), the tank nominal volume shall not be more than 25 kg.	The sprayer tank has a tank nominal volume of 20 L. It has a weight of 30.2 kg with water.
17. For compression knapsack sprayer, the spray tank volume shall exceed the nominal volume by at least 25%. For other sprayers (i.e., lever-operated, engine-driven, electric motor-driven, and dual), the spray tank volume shall exceed the nominal volume by at least 5%.	Not applicable
18. The sprayer shall remain upright and stable when placed on a level surface. It shall be easy for the user to lift the sprayer at full tank and its components onto the back.	a. The sprayer remained upright and stable when placed on a level surface during the test operation. b. The operator was able to carry the sprayer filled with water at tank nominal volume through the use of the straps provided.
19. The spray tank shall have a minimum capacity of 10 L.	The sprayer tank has a nominal volume of 20 L.
20. The nominal volume shall be specified in whole liter (L).	Nominal volume is specified in whole liter (L).
21. The sprayer shall contain a minimum of two filters, which are made of corrosion resistant material that allows for easy cleaning, maintenance, and/or replacement.	The sprayer has two (2) filters. One filter at the tank filler and one filter inside the tank.
22. The design of the spray tank inlet strainer shall be such that it allows rapid filling of the tank without splashing. The spray tank inlet strainer shall have at least 16 holes per square centimeter (mesh 16/cm ²).	Splashing was not observed during the filling of the tank. The inlet strainer has 256 holes per square centimeter.
23. The sprayer shall be filled through the spray tank inlet without spillage.	Spillage was not observed during the filling of the tank.
24. The lid shall allow easy and secure fitting by the gloved hand of an operator and shall provide an effective seal. There should be a provision for chain, string, or any form of attachment of the lid to the sprayer tank to avoid misplacement of the lid.	a. The lid can be tightened with a gloved hand and is provided with a seal. b. There is no provision to avoid lid misplacement.
25. Any air vent in the lid or tank shall limit the escape of spray liquid to 5 mL if the sprayer is completely inverted for five minutes.	About 424 mL spilled from the lid after complete inversion.

OBSERVATIONS (Cont'n)

26. The hose should be made of either rubber or synthetic material. If made of rubber, it shall have one or more plies of fiber reinforcement.	The hose is made of synthetic material.
27. Hoses shall be retained on connectors and couplings preferably by clamps or clips of the worm drive type. Threaded connections may be of any design provided that the strength and size permit liquid tight joints to be made by thumb pressure at the highest operating pressure of the sprayer.	Worm drive-type clamps or clips and threaded connectors are used to connect the hoses. The strength and size of the threaded connections were not verified due to limited technical capabilities.
28. To allow free handling of the spray lance, the length of the hose from the hose nipple of the spray tank to that of the spray lance hand grip shall be at least 1200 mm.	The length of the hose from the hose nipple of the spray tank is 1225 mm.
29. The spray lance shall provide a sufficient distance from the operator to the spray. The length of the lance shall be at least 500 mm.	The lance provided is 1245 mm long.
30. The cut-off valve shall have a lock-on and lock-off feature.	The cut-off valve is knob-type which can be turned on/off by rotating the knob.
31. The cut-off valve shall not leak or break when subjected to leak test.	No leak was observed during the test operation.
32. At least two nozzles that can deliver two different spray qualities (e.g., fine, medium, or coarse) depending on the intended use, as characterized in Table 2 (Droplet size classification) of PNS/BAFS 331:2022, shall be provided.	One single orifice nozzle is provided. It has a medium spray quality.
33. The nozzle body should contain a filter, which can be cleaned, maintained, or replaced. It shall be provided with a standard thread to fit the lance.	A nozzle filter was not provided.
34. Adjustable strap/s shall be provided to carry the sprayer.	Adjustable straps are provided.
35. All materials used for the strap shall be durable.	The straps are made of non-absorbent cloth.
36. The length of strap shall be minimum of 425 mm.	The minimum length of the strap is 630 mm.
37. The sprayer strap, straphangers, or strap clip shall not fail or get damaged during operation when subjected to strap test.	The strap drop test was not conducted as decided by the test applicant.
38. A quick-release mechanism shall be provided for emergency purposes.	Not provided
39. The increase in mass of straps after defined immersion in water shall not exceed 30% of the dry mass.	The straps' mass increased by 24.12% after 2-minute immersion in water.

OBSERVATIONS (Cont'n)

40. All hose connections shall have the same diameter. They shall be interchangeable and provided with reusable clamps.	The hose connections are of the same diameter and are provided with reusable clamps.
41. All other external connectors shall be of standard thread size. They shall be designed to avoid a potential source of leakage.	a. All other external connectors are of standard thread size. b. There were no leaks on the threaded connections during the test operation.
42. Fasteners shall not penetrate the pressure chamber.	Fasteners do not penetrate the pressure chamber.
43. A pressure gauge, if provided, shall have an accuracy of +1% of the full scale.	A pressure gauge is not provided.
44. The gauge shall be in a position where it is clearly visible when spraying and shall be installed before the cut-off valve.	Not applicable; no pressure gauge was provided
45. A fluid level indicator, if provided, should be clearly marked and visible, at 1-liter graduation with maximum error of +10%.	The fluid level indicators are marked per liter and numbers, 5, 10, 15, and 20L. The maximum error is not indicated.
46. Engine-driven knapsack sprayer	
a. After the pressure test is conducted and after a duration of 30 s, as defined by PNS/BAFS 332:2022, the total volume of leakage shall: i. Have no leak when in upright position; ii. Not exceed 0.5 mL when in 45° position; and iii. Not exceed 5 mL when in horizontal position.	Not verified; pressure test was not conducted due to limited technical capability.
b. The amount of liquid remaining in the tank shall not exceed 50 mL after operation.	About 40 mL of water remained in the tank after the spraying operation.
c. Engine starting device shall be provided to allow starting of the electric motor without the need for separate and independent auxiliary assistance (e.g., belts or cables). If the electric motor is fitted with an electric starting device, two or more independent and dissimilar motions shall be required to engage the device. The sprayer shall be fitted with an electric motor-stopping device through which the engine can be brought to a full stop and that does not depend on sustained manual effort for its operation.	The gasoline engine can be started through a rope recoil starter. A stop button is provided.

OBSERVATIONS (Cont'n)

d. The load bearing part of the strap shall be at least 65 mm wide and shall have a length of at least 200 mm and a thickness of at least 10 mm.	The strap is 67mm wide, 12.25 mm thick, and with a minimum length of 280 mm.
e. Fuel inlet shall have a strainer. Fuel tanks shall have an air vent.	Not provided
f. Fuel-tank openings shall be at least 20 mm in diameter and the oil-tank opening (if any) shall be at least 15 mm in diameter.	The diameter of the fuel and oil tank openings are 28.75 mm and 16.24 mm, respectively.
g. The design of the fuel-tank assembly shall be such that no leakage occurs in any operating and transport position at normal operating temperature.	No leaks were observed during the test operation.
h. Pressure control (e.g., relief valve) shall prevent excessive pressure buildup on the sprayer and its components.	Pressure control is provided.
i. It should be positioned external to the tank. If positioned internally, there shall be means of adjustment from the outside of the tank.	The pressure control is located on top of the engine's stop button.
j. Insulation and protection devices shall be provided to minimize the possibility of inadvertent contact with any exposed element which may cause burns during mounting, dismounting, or operating the machinery. Legible and non-removable warning notices or signs made of heat resistant materials for hot parts shall be provided.	a. Hot surfaces of the engine are covered and are not accessible to the operator when the machine is worn. b. Warning notices are affixed to the engine, a part which may cause danger to the operator, in the form of stickers. The warning notices are visible to the operator and indicate the particular danger that the operator is at risk of (Figure 2)
k. Exhaust outlets shall be located to direct exhaust emissions away from the operator in the normal operating position.	The exhaust outlet directs the emissions away from the operator.
47. All parts of the engine which are under high voltage shall be insulated to avoid hazards or accidents.	There are no exposed parts under high voltage. Wirings are enclosed inside a flexible hose.
48. All parts of the engine which are under high voltage shall be insulated to avoid hazards or accidents.	There are no exposed parts under high voltage. Wirings are enclosed inside a flexible hose.
49. There shall be a provision of earmuffs or other ear protection device for the operator to use when 92 dB(A) is exceeded during operation.	The maximum noise level during the continuous running test operation was 95.1 dB(A).

OBSERVATIONS (Cont'n)

50. There shall be a provision of earmuffs or other ear protection device for the operator to use when 92 dB(A) is exceeded during operation.	The maximum noise level during the continuous running test operation was 95.1 dB(A). No earmuffs were provided.
51. Vibration reduction shall be an integral part of the design process, specifically taking into account measures at source. Technical measures such as isolators and resonating masses shall be used to isolate the vibration from the handle, when appropriate.	It was not verified if the machine's vibration was minimized/reduced because of limited technical capabilities to quantify and compare vibration levels during the test operation.
52. The sprayer shall include among others at least one set of protective nose and mouth mask and a pair of eye goggles to protect the operator against inhalation and eye irritation, respectively due to spray chemicals.	Not provided
53. The sprayer shall conform to the operator's body, shall have a weight distributed evenly, and shall have operating controls in reasonable locations and configuration.	a. The operator was able to carry the sprayer on the back. A foam lining at the back of the tank is provided. b. The sprayer's pressure control and the engine's throttle lever and stop button are located on the left side of the operator and below the spray tank.
54. No part of the outer surface shall entrap spilled liquid and there shall be no sharp edges or protrusions.	There are no sharp edges or protrusions on any part of the sprayer.
55. All valve parts shall be readily accessible for servicing and replacement.	Valve parts are accessible for replacement and servicing.
56. There should be a convenient clip or any holder for the lance when not in use.	Not provided
57. The cut-off valve shall be easy to use and comfortable to operate.	The sprayer utilized a knob-type cut-off valve which requires to be rotated to turn on/off the spray.
58. The cut-off valve's handle grip shall be comfortable to hold and easy to use.	The sprayer utilized a knob-type cut-off valve which also serves as its handle.
59. Engine or electric motor-driven sprayers shall be constructed in such a manner that access to power-driven components such as pulleys, shafts, gears, and flywheels, and to drive belts and chains is prevented. For openings (e.g., covers and guards) the safety distances shall be in accordance with PNS/BAFS 330:2022 (Technical means for ensuring safety – Guidelines).	Power-driven components are covered and are not accessible to the operator when the machine is worn.

OBSERVATIONS (Cont'n)

60. Other remarks	The test applicant decided to continue the test operation with only one nozzle.
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PURPOSE AND SCOPE OF TEST

Purpose	Performance Testing
Dates of Test	November 4, 17, and 18, 2024
Location of Test	Roberto C. Bautista Hall, UPLB, College, Los Baños, Laguna
Items Tested	Operator's Manual, Tilt and Inversion, Discharge Rate, Droplet Size Analysis, Continuous Running, Durability of the Strap, Strap Absorbency, Drop Test, Fuel Consumption, Noise Level, Speed of Components, and Number of Operators

METHODS OF TEST

The engine-operated knapsack sprayer was tested based on PNS/BAFS 332:2022 Agricultural Machinery – Knapsack Sprayer – Methods of Test.

DESCRIPTION OF THE MACHINE

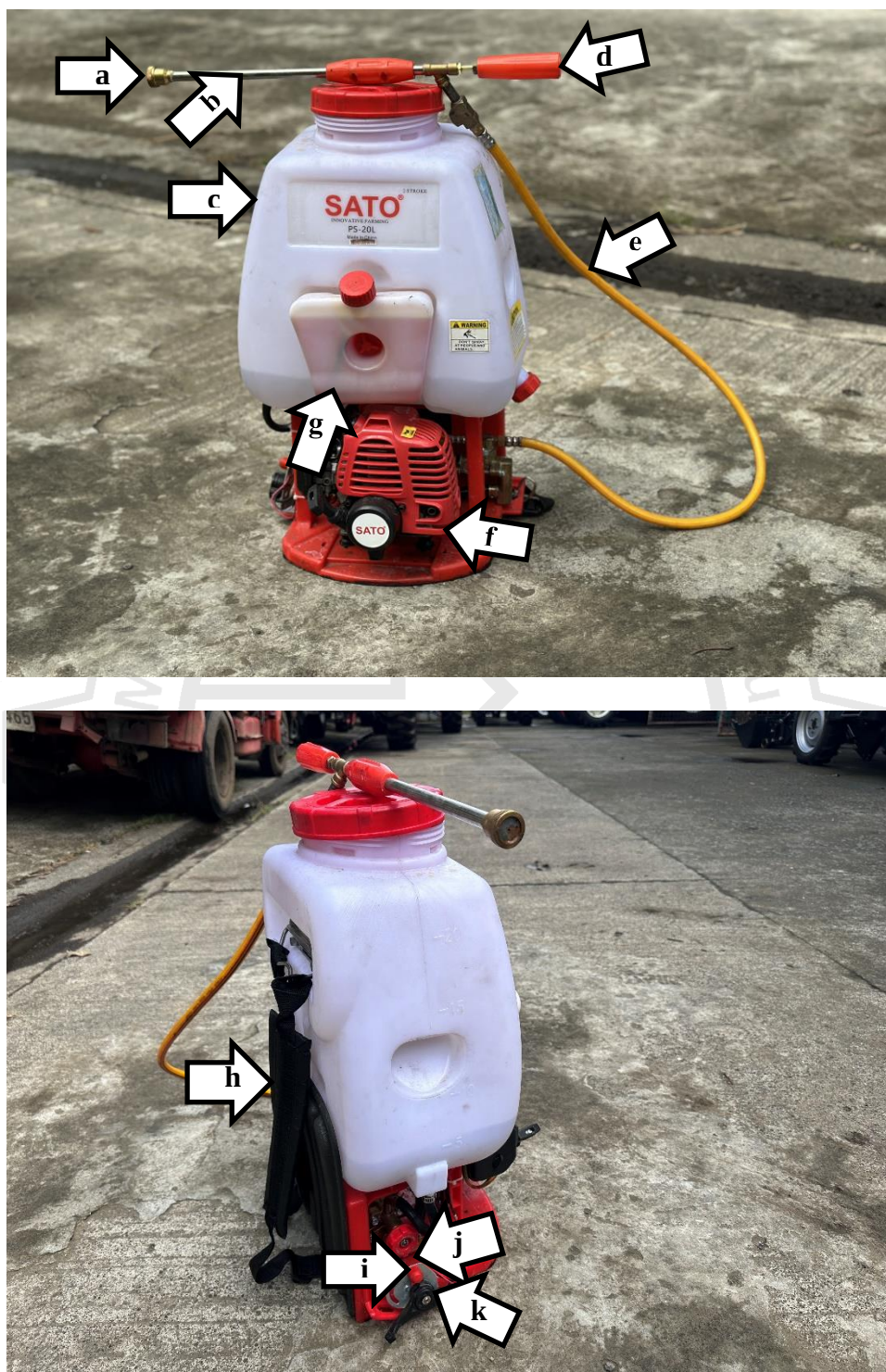


Figure 1. Parts of the SATO PS-20L-TU26 Knapsack Sprayer include the following:
(a) Nozzle, (b) Lance, (c) Spray tank, (d) Cut-off valve, (e) Delivery hose,
(f) Engine, (g) Fuel tank, (h) Straps, (i) Engine stop button,
(j) Pressure control knob, and (k) Throttle lever.

SPECIFICATIONS

Item		Manufacturer's Specification ^a	AMTEC Verification
1	Dimension (in transport mode), mm		
1.1	Length	400	390
1.2	Width	330	285
1.3	Height	650	605
2	Weight, kg		
2.1	Without water	9	8
2.2	With water (tank capacity)	ND	30.2
3	Working Pressure, kPa		
3.1	Normal	1000	1000 ^a
3.2	Maximum	4000	2234 ^b
4	Tank		
4.1	Volume, L		
4.1.1	Nominal	25	20.00
4.1.2	Total	ND	21.63
4.2	Dimension, mm		
4.2.1	Length	ND	395
4.2.2	Width	ND	390
4.2.3	Height	ND	240
4.3	Diameter of filler hole, mm	ND	145
4.4	Material	ND	Plastic
5	Piston Assembly		
		ND	NA

^a The data were obtained from the operator's manual and specification sheet provided by the test applicant.

^b Based on discharge test.

Note: NA – Not Applicable (The sprayer is engine-driven), ND – No Data.

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification ^a	AMTEC Verification
6	Pressure Chamber		
6.1	Volume, L	ND	NM ^b
6.2	Shape	ND	NM ^b
6.3	Location	ND	NM ^b
6.4	Material	ND	NM ^b
7	Cut-off Valve		
7.1	Type	ND	Knob
7.2	Material	ND	Metal with plastic handle
7.3	Lock -on/Lock-off feature	ND	None
8	Delivery Hose		
8.1	Length, mm	ND	1245
8.2	Inside diameter, mm	ND	NM ^b
8.3	Thickness, mm	ND	NM ^b
8.4	Clamping device	ND	Clip
8.5	Material	ND	Metal
9	Lance		
9.1	Shape	ND	Cylinder
9.2	Length, mm	ND	1245
9.3	Inside diameter, mm	ND	NM ^b
9.4	Material	ND	204 SS ^c
10	Grip (lance)		
10.1	Shape	ND	Oblong
10.2	Size, D × L, mm	ND	30.39 × 110
10.3	Material	ND	Plastic

^a The test applicant did not submit complete specifications of the machine.

^b The component to be measured for the required parameters is not accessible.

^c Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer; see Table 9 for the elemental composition of the material.

Note: NA – Not Applicable, ND – No Data, NM – Not Measured

SPECIFICATIONS (Cont'n)

Item		Manufacturer's Specification^a	AMTEC Verification
11	Nozzle		
11.1	Type	ND	Single orifice
11.2	Material	ND	430/440 SS ^b
12	Strainer/Filter		
12.1	Basket strainer		
12.1.1	Size of mesh, hole/cm ²	ND	169
12.1.2	Material	ND	Plastic
12.2	Inlet-strainer (pump)		
12.2.1	Size of mesh, hole/cm ²	ND	256
12.2.2	Material	ND	Plastic
12.3	Inlet-strainer (lance)	ND	NA
12.4	Nozzle filter	ND	NA
13	Straps		
13.1	Material	ND	Non-absorbent cloth
13.2	Dimension, mm		
13.2.1	Length	ND	860
13.2.2	Width	ND	38
13.2.3	Thickness	ND	1.20

^a The test applicant did not submit complete specifications of the machine.

^b Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer; see Table 9 for the elemental composition of the material.

Note: NA – Not Applicable (No filter was provided on the parts of the sprayer), ND – No Data.

RESULTS

Tilt and inversion

Table 1. Tilt and inversion test of the SATO PS-20L-TU26 Knapsack Sprayer.

Position	Observation
Tilted at 45° angle front side	No leak observed
Tilted at 45° angle back side	No leak observed
Tilted at 45° angle right side	No leak observed
Tilted at 45° angle right side	No leak observed
Horizontal position front side	Leak observed
Horizontal position back side	Leak observed
Horizontal position right side	Leak observed
Horizontal position right side	Leak observed

Discharge test

Table 2. Discharge test of the SATO PS-20L-TU26 Knapsack Sprayer.

Pressure, kPa	Noise level, dB(A)	Engine speed, rpm	Fuel consumption, L/h	Discharge rate, L/min	Water power, kW	Input power, kW	System efficiency, %
2234	95.2	6941	0.031	6.15	0.229	0.304	75.54
1558	93.4	6961	0.012	4.20	0.109	0.115	95.24
883	94.1	7233	0.018	3.07	0.045	0.179	25.26
207	94.0	7538	0.020	1.57	0.005	0.195	2.78

Droplet Analysis

Table 3. Spray droplet size, distribution, and volume for SATO PS-20L-TU26 Knapsack Sprayer.

Water SP position ^a	Single Orifice Nozzle
Volume median diameter, μm	236.00
Droplet distribution, spray droplets/cm ²	9.07
Deposition, $\mu\text{L}/\text{cm}^2$	0.338
Average DV 5, μm	236
Spray quality ^b	Medium

^a Points 1 to 3 were taken at intervals of 250 mm from the center of spray delivered at a distance of 1000 mm and height of 1000 mm.

^b Based on the American Society of Agricultural and Biological Engineers (ASABE). 2020. Droplet size classification (ASABE S-572.1).

RESULTS (Cont'n)**Spray angle**

Table 4. Spray angle of the SATO PS-20L-TU26 Knapsack Sprayer.

Type of nozzle	Single orifice nozzle
Spray angle, °	7.5

Continuous running test

Table 5. Continuous running test of the SATO PS-20L-TU26 Knapsack Sprayer using Nozzle 1.

Time elapsed, h	Operating pressure, kPa
0	2420.15
1	2275.35
2	3695.72
3	2882.11
4	3399.24
5	3481.98
Total time, h	5

Strap test

Table 6. Strap test of the SATO PS-20L-TU26 Knapsack Sprayer.

Item	Data
Height of lift, mm	200
Number of times dropped	4
Observations	Metal clips on straps failed after 4th attempt (Figure 4)

RESULTS (Cont'n)**Strap absorbency test**

Table 7. Strap absorbency test of the SATO PS-20L-TU26 Knapsack Sprayer.

Item	Data
Ambient conditions	
Temperature, °C	28.7
Relative humidity, %	75.6
Initial weight, g	72.05
Final weight, g	89.43
Absorbency rate, %	24.12

Drop test

Table 8. Drop test of the SATO PS-20L-TU26 Knapsack Sprayer.

Item	Data
Height of lift, mm	600
Number of times dropped	1
Observations	No deformations were observed.

RESULTS (Cont'n)

Table 9. Elemental composition of the different components of the SATO PS-20L-TU26 Knapsack Sprayer.

Element	Percent Composition^a	
	Nozzles	Lance
Iron	82.807	78.857
Chromium	16.358	0.152
Manganese	0.450	0.135
Nickel	0.160	0.830
Copper	0.092	0.536
Vanadium	0.082	0.051
Tungsten	0.024	<LOD
Molybdenum	0.019	0.074
Lead	0.016	<LOD
Tin	0.014	0.168
Titanium	<LOD	<LOD
Cobalt	<LOD	<LOD
Zirconium	<LOD	<LOD
Niobium	<LOD	<LOD
Tantalum	<LOD	<LOD
GRADE	430/440 SS	204 SS

^a Determined using the BRUKER S1 TITAN 800 X-ray Fluorescence Analyzer.

Note: LOD – Limit of Detection

OBSERVATIONS

Figure 2. Nameplate, markings, labeling, and safety warnings on the SATO PS-20L-TU26 Knapsack Sprayer.



Figure 3. Spillage at different positions during the tilt and inversion test of the PS-20L-TU26 SATO Knapsack Sprayer.

OBSERVATIONS



Figure 4. Metal clip strap failure after 4th attempt of the strap drop test.

OBSERVATIONS



Figure 5. Conformity of the sprayer to the back of the operator during testing.

Tested by:


PAUL JOHN S. DIZON
Test Engineer


JOHN PAUL A. PALILLO
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Verified by:


MELVIN T. BALDRIAS
Test Engineer

Approved for Release:


ARTHUR L. FAJARDO
AMTEC Director

30 NOV 2024
Date

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